

(*) Application of DE...

Q. A chain coiled up near the edge of a smooth table starts to fall over the edge. The velocity 'v' when a length 'x' has fallen is given by

$$xv \frac{dv}{dx} + v^2 = gx. \quad \text{show that when}$$

$$g = 32, \quad v = 8\sqrt{\frac{x}{3}}$$

Ans Given : velocity $\rightarrow v$
 $\hookrightarrow$ when a length 'x'

$$\hookrightarrow xv \frac{dv}{dx} + v^2 = gx \quad \text{--- (1)}$$

$\hookrightarrow$ this is a differential eq. in v & x

$\rightarrow$ Divide both sides by xv in (1)

$$(1) \Rightarrow \frac{dv}{dx} + \frac{v}{x} = \frac{g}{v} \quad \left. \vphantom{\frac{dv}{dx} + \frac{v}{x} = \frac{g}{v}} \right\} \text{1st order form}$$

$\rightarrow$ Multiply both sides by v

$$\Rightarrow v \frac{dv}{dx} + \frac{v^2}{x} = g \quad \text{--- (2)}$$

(X) $\rightarrow$ substitute : $v^2 = t$

$$\Rightarrow 2v \cdot \frac{dv}{dx} = \frac{dt}{dx} \Rightarrow \frac{dv}{dx} = \frac{1}{2v} \frac{dt}{dx}$$

$\hookrightarrow$ substitute in (2)

$$\Rightarrow \frac{1}{2v} \left(\frac{dt}{dx} \right) + \frac{t}{x} = g$$

$\rightarrow$ Multiply through by 2

$$\Rightarrow \frac{dt}{dx} + \frac{2t}{x} = 2g \quad \text{--- (3)}$$

→ this is a linear differential eq. in t

(*) standard Form :

comparison $\frac{dt}{dx} + P(x)t = Q(x)$

$$\therefore \text{(3)} \Rightarrow P = \frac{2}{x} \quad \& \quad Q = 2g$$

→ Integrating factor : $\rightarrow e^{\int P dx}$

$$\Rightarrow e^{\int \frac{2}{x} dx} = e^{2 \ln x} = x^2$$

$\therefore$ Multiply (3) by x^2

$$x^2 \frac{dt}{dx} + 2xt = 2gx^2$$

$$\Rightarrow \frac{d}{dx}(tx^2) = 2gx^2$$

→ Integrate both sides (w.r.t. x)

$$tx^2 = 2g \int x^2 dx$$

$$tx^2 = 2g \frac{x^3}{3} + C$$

→ Now substitute back $t = v^2$

$$\Rightarrow v^2 x^2 = 2g \frac{x^3}{3} + C \quad \text{--- (4)}$$

→ Apply the initial condition

→ When $x = 0$, the chain has just started to move $\therefore v = 0$

(3)

$$\therefore (4) \Rightarrow 0 = 0 + C \Rightarrow C = 0$$

$$\therefore (4) \Rightarrow v^2 x^2 = \frac{2g x^3}{3}$$

Divide both sides by x^2

$$\Rightarrow v^2 = \frac{2gx}{3}$$

→ Now substitute $g = 32$

$$v^2 = \frac{2(32)x}{3} = \frac{64x}{3}$$

→ take sq. root.

$$\Rightarrow v = \sqrt{\frac{64x}{3}} = 8\sqrt{\frac{x}{3}}$$

$$\therefore \boxed{v = 8\sqrt{\frac{x}{3}}} \text{ Ans}$$

Q2. Solve $\frac{di}{dt} + \frac{R}{L}i = \frac{E}{L}$ for the case in which the circuit has initial current i_0 at time $t=0$ & e.m.f impressed is given by $E = E_0 e^{-kt}$

Ans Given : a linear first-order O.D.E for current $i(t)$ when the impressed emf is $E(t) = E_0 e^{-kt}$

$$\rightarrow \frac{di}{dt} + \frac{R}{L}i = \frac{E}{L} = E_0 e^{-kt} \quad \text{--- (1)}$$

$$\text{Also } i(0) = i_0$$

→ Comparison with std. form:

$$\frac{di}{dt} + P(t)i = Q(t)$$

$$\textcircled{1} \Rightarrow P(t) = \frac{R}{L} \text{ (const)} ; Q(t) = \frac{E_0}{L} e^{-kt}$$

→ Integrating factor: $e^{\int P dt}$
 (I.F) $\Rightarrow e^{\int R/L dt}$
 $\Rightarrow e^{R/L t}$

→ Multiply $\textcircled{1}$ by the I.F

$$e^{(R/L)t} \frac{di}{dt} + \frac{R}{L} e^{(R/L)t} i = \frac{E_0}{L} e^{(R/L)t} e^{-kt}$$

$$\Rightarrow \frac{d}{dt} (i e^{(R/L)t}) = \frac{E_0}{L} (e^{((R/L)-k)t})$$

→ Integrate both sides (w.r.t. 't')

$$\Rightarrow i e^{(R/L)t} = \int \frac{E_0}{L} e^{((R/L)-k)t} dt$$

$$\Rightarrow i e^{(R/L)t} = \frac{E_0}{L} \frac{e^{((R/L)-k)t}}{(R/L)-k} + C \quad \textcircled{2}$$

* → Simplify the coefficient

$$\frac{R}{L} - k = \frac{R - Lk}{L}$$

$$\Rightarrow \frac{E_0}{L} \cdot \frac{1}{(R/L) - k} = \frac{E_0}{L} \cdot \frac{L}{R - kL} = \frac{E_0}{R - kL}$$

$$\therefore (2) \Rightarrow i e^{(R/L)t} = \frac{E_0}{R - kL} \cdot e^{((R/L) - k)t} + C \quad (3)$$

$$\text{Now: } e^{((R/L) - k)t} = e^{(R/L)t} \cdot e^{-kt}$$

→ On (3) Divide both sides by $e^{(R/L)t}$

$$\Rightarrow i(t) = \frac{E_0}{R - kL} e^{-kt} + C e^{-(R/L)t} \quad (4)$$

→ Apply initial condition

$$\text{At } t = 0, i(0) = i_0 \therefore$$

$$i_0 = \frac{E_0}{R - kL} + C \Rightarrow C = i_0 - \frac{E_0}{R - kL}$$

$$\therefore (4) \Rightarrow i(t) = \frac{E_0}{R - kL} e^{-kt} + \left(i_0 - \frac{E_0}{R - kL} \right) e^{-(R/L)t} \quad \underline{\text{Ans}}$$

→ In a circuit containing inductance L , resistance R & voltage E is given by $E = Ri + L \frac{di}{dt}$ if $L = 640 \text{ H}$, $R = 250 \Omega$, $E = 500 \text{ V}$ & $i = 0$ when $t = 0$. Find the time that elapses before the current reaches 90% of its max. value.

Ans Given: $E = Ri + L \frac{di}{dt}$ [circuit law Kirchhoff + inductor voltage]

$$L = 640 \text{ H}, R = 250 \Omega; E = 500 \text{ V}$$

$$i(0) = 0$$

→ Rearrange : to form std. 1st order linear DDE

$$\Rightarrow \frac{di}{dt} + \frac{R}{L} i = \frac{E}{L} \quad \text{--- ①}$$

• Comparison with the std. form : $\frac{di}{dt} + Pi = Q$

$$\text{①} \Rightarrow P = \frac{R}{L} \quad \text{and} \quad Q = \frac{E}{L}$$

• I.F : $e^{\int P dt} = e^{\int R/L dt} = e^{\frac{R}{L}t}$

• Multiply ① by I.F

$$\Rightarrow e^{(R/L)t} \frac{di}{dt} + \frac{R}{L} e^{(R/L)t} i = \frac{E}{L} e^{(R/L)t}$$

$$\Rightarrow \frac{d}{dt} (i e^{(R/L)t}) = \frac{E}{L} e^{(R/L)t}$$

→ Integrate both side (w.r.t. 't')

$$i e^{(R/L)t} = \int \frac{E}{L} e^{(R/L)t} dt$$

$$\Rightarrow i e^{(R/L)t} = \frac{E}{L} \cdot \frac{e^{(R/L)t}}{(R/L)} + C$$

$$\Rightarrow i e^{(R/L)t} = \frac{E}{R} e^{(R/L)t} + C$$

→ Divide by $e^{(R/L)t}$

$$\Rightarrow i(t) = \frac{E}{R} + C e^{-(R/L)t}$$

* → Initial Condⁿ. App.

$$i(0) = 0$$

$$\Rightarrow i(0) = \frac{E}{R} + C e^{-(R/L)(0)}$$

$$\Rightarrow 0 = \frac{E}{R} + C \Rightarrow C = -\frac{E}{R}$$

$$\therefore i(t) = \frac{E}{R} [1 - e^{-(R/L)t}]$$

Ans

→ Assume max value of $i = I$ at $t \rightarrow \infty$

$$\therefore i(\infty) = \frac{E}{R} [1 - e^{-(R/L)\infty}] \quad (e^{-\infty} = 0)$$

$$I = \frac{E}{R} [1 - e^{-(R/L)\infty}]$$

$$\therefore I = \frac{E}{R} [1 - 0] \Rightarrow I = \frac{E}{R}$$

→ When current reaches 90% of its max value

$$\hookrightarrow i = 90\% I = \frac{9}{10} \frac{E}{R}$$

$$\therefore \rightarrow \frac{9}{10} \frac{E}{R} = \frac{E}{R} (1 - e^{-(R/L)t}) \quad ?$$

$$\Rightarrow e^{-(R/L)t} = 1 - \frac{9}{10} = \frac{1}{10} \Rightarrow e^{R/L t} = 10$$

take log on both sides

$$\Rightarrow \log e^{R/L t} = \log 10 \Rightarrow \frac{R}{L} t = \log 10$$

$$\Rightarrow t = \frac{L}{R} \log 10 = \frac{640}{250} \log 10$$

$$= \boxed{5.8946 \text{ sec}}$$

Ans

Q4. The charge q on the plate of a condenser of capacity C charged through a resistance R by the steady voltage V satisfies the differential eq

$$R \cdot \frac{dq}{dt} + \frac{q}{C} = V \text{ if } q=0 \text{ at } t=0 \text{ show}$$

that $q = CV(1 - e^{-t/RC})$. Find also the current flowing into the plate.

Q5. An eq. in the theory of stability of an aeroplane is $\frac{dv}{dt} = g \cos \alpha - kv$, v being velocity & g, k being constants, it is observed that at $t=0$ $v=0$. Solve the eq. completely.

Variation of Parameters

Method for solving 2nd-order linear D.E

→ To Find : Particular solution y_p [to a non-homogeneous linear diff. Eq]

→ Only : 2nd-order linear D.E

→ $f(D)y = X$ ← operator form of a linear D.E
 $D \leftarrow \frac{d}{dx}$

$f(D)$ is some polynomial in D ; y is the unknown function, X is a given function (non-homog. term)

→ Complementary function : ← the solⁿ to the homog. part. $f(D)y = 0$

$$y_c = C_1 y_1 + C_2 y_2$$

y_1 & y_2 are linearly indep.

→ Wronskian : ← helps check y_1 & y_2 are linearly indep. or not.

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

if $W \neq 0$ ← they are independent

→ Particular solⁿ : assumed in this form

$$y_p = u y_1 + v y_2$$

→ where u and v are functions to be determined: (not constants)

→ Formulas for u & v

$$u = - \int \frac{y_2 X}{W} dx$$

$$v = \int \frac{y_1 X}{W} dx$$

→ Total solution :

$$y = y_c + y_p$$

← gen. solⁿ
c.b. + part. sol

Example ① Solve $(D^2 + D)y = \frac{1}{1+e^x}$

Ans $(D^2 + D)y = \frac{1}{1+e^x}$ ← non-homo.

→ solve the homo. eq

$$(D^2 + D)y = 0$$

$$\Rightarrow D(D+1)y = 0$$

→ characteristic eq.

$$m(m+1) = 0 \Rightarrow m = 0, m = -1$$

$$\therefore y_c = c_1 e^{0x} + c_2 e^{-x} = c_1 + c_2 e^{-x}$$

$$y_1 = 1$$

$$y_2 = e^{-x}$$

→ Find the Wronskian

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} 1 & e^{-x} \\ 0 & -e^{-x} \end{vmatrix} = -e^{-x} \neq 0$$

$\therefore y_1$ & y_2 are indep.

→ identity X

↳ from the original eq: $X = \frac{1}{1+e^x}$

→ Find u & v

$$\rightarrow u = -\int \frac{y_2 x}{w} dx \quad ; \quad v = \int \frac{y_1 x}{w} dx$$

$$\therefore u = -\int \frac{e^{-x} \cdot \frac{1}{1+e^x}}{-e^{-x}} dx \quad \Bigg| \quad = \int \frac{1 \cdot \frac{1}{1+e^x}}{-e^{-x}} dx$$

$$u = \int \frac{1}{1+e^x} dx$$

$$= -\int \frac{e^x}{1+e^x} dx$$

multiply N^r & D^r by e^{-x}

$$u = \int \frac{e^{-x}}{1+e^{-x}} dx$$

$$\therefore u = \log(1+e^{-x})$$

$$v = -\log(1+e^x)$$

→ sub. in $y_P = u y_1 + v y_2$

$$\therefore y_P = + \left[-\log(1+e^x) \right] (e^{-x})$$

$$\therefore y_P = \log(1+e^{-x}) - e^{-x} \log(1+e^x)$$

→ Final solⁿ: $y = y_c + y_P$

$$\therefore y = (C_1 + C_2 e^{-x}) + \left[\log(1+e^{-x}) - e^{-x} \log(1+e^x) \right]$$

② Use the method of variation of parameters to solve

④

$$y'' + 3y' + 2y = e^{e^x}$$

Ans Convert to operator form

$$\therefore (D^2 + 3D + 2)y = e^{e^x}$$

$$(D = \frac{d}{dx})$$

→ Solve the homo. eq.

$$D^2 + 3D + 2 = 0$$

$$\text{Charac. eq: } m^2 + 3m + 2 = 0$$

$$\therefore m = -1 \text{ \& } -2$$

$$\therefore y_c = c_1 e^{-x} + c_2 e^{-2x}$$

$$\therefore y_1 = e^{-x} \text{ \& } y_2 = e^{-2x}$$

→ Find the Wronskian

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{-x} & e^{-2x} \\ -e^{-x} & -2e^{-2x} \end{vmatrix} = -2e^{-3x} + e^{-3x} = -e^{-3x} \neq 0 \therefore \text{indep.}$$

→ Identify x $\therefore x = e^{e^x}$

→ Find u & v

$$\begin{aligned} \therefore u &= -\int \frac{y_2 x}{W} dx \\ &= -\int \frac{e^{-2x} \cdot e^{e^x}}{-e^{-3x}} dx \\ &= \int e^x \cdot e^{e^x} dx \end{aligned}$$

$$e^x = t \Rightarrow e^x dx = dt$$

$$\Rightarrow \int e^t dt = e^t = e^{e^x}$$

$$\begin{aligned} v &= \int \frac{y_1 x}{W} dx \\ &= \int \frac{e^{-x} \cdot e^{e^x}}{-e^{-3x}} dx \\ &= -\int e^{2x} \cdot e^{e^x} dx \\ &= -\int e^x \cdot e^x \cdot e^{e^x} dx \\ &= -\int e^t \cdot t dt \\ &= -\left[(t)(e^t) - (1)e^t \right] = -e^{e^x} + e^{e^x} \end{aligned}$$

$\therefore \rightarrow$ sub in $y_p = u y_1 + v y_2$

$$\begin{aligned} \therefore y_p &= e^{e^x}(e^{-x}) + [-e^x e^{e^x} + e^{e^x}](e^{-2x}) \\ &= \cancel{e^{e^x} e^{-x}} - \cancel{e^{-x} e^{e^x}} + e^{e^x} \cdot e^{-2x} \end{aligned}$$

$\rightarrow$ Final solⁿ:

$$\therefore y = y_c + y_p$$

$$\therefore y = (c_1 e^{-x} + c_2 e^{-2x}) + \underline{\underline{e^{e^x} \cdot e^{-2x}}}$$

③ Solve $(D^2+1)y = \operatorname{cosec} x \cot x$ using variation of parameters

Ans Homog. part:

$$(D^2+1)=0 \quad \therefore \text{char. eq}^n: m^2+1=0$$

$$\Rightarrow m = \pm i$$

$$\Rightarrow m = a \pm ib \text{ form}$$

$$a=0, b=1$$

$$\therefore y_c = c_1 \cos x + c_2 \sin x$$

$$\therefore y_1 = \cos x \text{ and } y_2 = \sin x$$

$$\begin{aligned} \rightarrow \text{Find Wronskian: } W &= \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} \\ &= \cos^2 x + \sin^2 x = 1 \neq 0 \end{aligned}$$

$\therefore$ indep.

$$\rightarrow \text{Identify } X \Rightarrow X = \operatorname{cosec} x \cot x$$

(6)

→ Find u & v

$$\begin{aligned}\therefore u &= -\int \frac{y_2 x}{w} dx \\ &= -\int \sin x \operatorname{cosec} x \cot x dx \\ &= -\int \frac{\cos x}{\sin x} dx \\ &= -\log(\sin x)\end{aligned}$$

$$\begin{aligned}v &= \int \frac{y_1 x}{w} dx \\ &= \int \cos x \operatorname{cosec} x \cot x dx \\ &= \int \frac{\cos x}{\sin x} \cdot \cot x dx \\ &= \int \cot^2 x dx \\ &= \int (\operatorname{cosec}^2 x - 1) dx \\ &= -\cot x - x\end{aligned}$$

→ Subs in $y_p = uy_1 + vx_2$

$$\therefore y_p = -\log(\sin x) \cos x - (\cot x + x) \sin x$$

∴ Final solⁿ : $y = y_c + y_p$

④ $(D^2 - 4D + 4)y = e^{2x} \sec^2 x$

⑤ $\frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 6y = e^{-2x} \sec^2 x (1 + 2 \tan x)$

⑥ $\frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 6y = e^{-2x} \sec^2 x (1 + 2 \tan x)$